Aperture Array-Based Self-Supervised Hyperspectral Super-Resolution Imaging

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***Abstract*—Current hyperspectral imaging (HSI) technologies struggle with prolonged acquisition time, bulky system design, and high-computational complexity. We propose aperture array-based self-supervised hyperspectral super-resolution (SR) imaging, a novel computational HSI framework that integrates an aperture array optical structure with a physics-driven, self-supervised deep learning model. The aperture array enables spectral division while leveraging aperture parallax for spatial SR, achieving high spectral fidelity and spatial SR in a compact form factor. To fully exploit the imaging characteristics of the proposed system, we design a divide-and-conquer self-supervised hyperspectral SR imaging (DSHSI) algorithm, which is data-efficient, noniterative, and free of large-scale ground-truth datasets. Unlike existing hyperspectral restoration methods that suffer from long inference times and poor generalization, DSHSI integrates physics-driven priors into a self-supervised learning framework, enabling robust performance on unseen scenes and out-of-distribution degradations. DSHSI jointly performs denoising, SR, and deblurring in a unified manner, reducing runtime by 25×, FLOPs by 9×, and model parameters by 3.6× than DualSR on public remote sensing datasets, while achieving a 1.9 dB PSNR gain, enabling fast and efficient hyperspectral reconstruction. In the optical experiments, it achieves the best NIQE and the spectral distortion metric that is closely aligned with the spectral response of observed HSI data. Our code is publicly available at https://github.com/THUHoloLab/DSHSR**

***Index Terms*—Aperture array, divide-and-conquer self-supervised learning, hyperspectral super-resolution (SR) imaging.**

I. Introduction

As THE fingerprint of matter, hyperspectral imaging (HSI) provides a powerful tool for analyzing the composition and concentration of substances. Unlike traditional imaging and multispectral imaging, which capture limited spectral bands, HSI acquires a vast amount of continuous narrowband spectral information and spatial intensity information. This capability has driven advancements across diverse fields, including precision agriculture and environmental monitoring [1], atmospheric remote sensing [2], and biomedical imaging and disease diagnostics [3]. Despite decades of progress, the commercialization of HSI remains in its early stages. Key challenges, such as high system costs, limited portability, and the computational burden of high-dimensional data processing, have constrained its widespread adoption.Existing HSI technologies can be categorized into scanning-based and snapshot-based approaches [7]. Scanning methods, including whiskbroom, pushbroom, and staring imaging, sequentially acquire spectral information across multiple measurements, making them impractical for dynamic scenes. In contrast, snapshot-based techniques employ compressive sampling to map the 3-D data onto a 2-D sensor array, enabling the capture of both spectral and spatial information in a single measurement, making them the most promising HSI approach currently available [8]. Coded aperture snapshot spectrometer utilizes binary coding [9], color coding [10], or random spectral coding [11] to achieve compressive sampling, and then reconstruct the full-resolution spectral cube using compressive sensing algorithms combined with data sparsity priors. However, the optical path length of this spectrometer is inversely proportional to its spectral resolution. Diffraction-based snapshot HSI leverages diffractive optical elements to replace traditional dispersive elements, coded aperture, relay lens, and imaging lens, achieving system miniaturization, but often at the cost of reduced image quality [12], [13]. Recent advances in metasurface-based on-chip spectrometers offer an ultracompact solution [14]. The spectral modulation manners of metasurfaces can be further categorized into broadband spectral modulation [15] and narrowband spectral modulation [16]. Alternatively, the thin-film deposition techniques can also be employed to achieve spectral filtering, such as in multispectral cameras using pixel-level narrowband filter arrays [17], [18]. To further improve light throughput, broadband HSI employs broadband filter arrays combined with computational reconstruction [19]. However, existing broadband decoding algorithms suffer from poor generalization, introducing spectral distortions when applied to unseen scenes. To overcome these limitations, this paper proposes a novel computational HSI technique—aperture array-based self-supervised hyperspectral super-resolution (SR) imaging—aimed at achieving HSI with high spatiotemporal resolution while maintaining a compact, fabrication-friendly design. Our approach leverages spatial multiplexing, using apertures as fundamental units to capture spatial information in individual narrowband spectral channels, while an array configuration simultaneously acquires spatial information across multiple bands. The physical constraints, such as sensor size limitations, energy

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loss from narrowband filtering, and bandwidth restrictions in high-speed data acquisition, often result in hyperspectral images with insufficient spatial resolution, hindering precise spectral analysis. To address this, our method exploits aperture parallax-induced subpixel shifts, fusing complementary spatial information across spectral bands through an SR reconstruction algorithm to improve spatial resolution.

Existing hyperspectral SR algorithms can be broadly classified into two categories: auxiliary hyperspectral image SR and single hyperspectral image SR. The former relies on a high-resolution (HR) color image or multispectral image (MSI) to provide fine-grained scene details [20], [21], [22], making it not directly applicable to the system proposed in this paper. The latter, single hyperspectral image SR, enhances spatial resolution by exploiting complementary spatial information across different spectral bands, relying solely on the observed hyperspectral image. However, current optimization-based single hyperspectral image SR methods require complex prior constraints to solve the ill-posed high-dimensional image reconstruction problem, such as low-rank and sparse priors in the 3-D tensor domain [23], [24], resulting in prolonged reconstruction time. Although existing deep learning-based single hyperspectral image SR methods significantly improve reconstruction speed, they primarily rely on supervised learning [4], [5], [25], [26], [27], [28], leading to insufficient spectral fidelity in the reconstructed images and a lack of physical model interpretability. Consequently, the limitations of low reconstruction quality, slow inference speed, and poor generalization ability hinder the widespread adoption of snapshot computational HSI systems.

To address these challenges, this paper formulates a physics-driven model for single hyperspectral image SR based on an aperture array, incorporating the statistical characteristics of observed hyperspectral images to guide the design of a self-supervised hyperspectral SR framework. Unlike supervised learning-based approaches that require large-scale, HR ground-truth data, the proposed model is entirely self-supervised, using only the input itself as the training target. This ensures strong generalization to unseen and real-world scenes. This paper proposes a lightweight attention-based dual-stage network architecture tailored for self-supervised learning, enabling high-quality reconstruction with a reduced number of parameters. As shown in Fig. 1, our method achieves superior results in both performance and computational efficiency compared with existing approaches. The network adopts an end-to-end, noniterative forward structure, significantly enhancing computational efficiency. Furthermore, a divide-and-conquer self-supervised learning framework is introduced, allowing different network modules to focus on specific tasks, thereby simultaneously achieving denoising, deblurring, and SR. Specifically, the fusion module is dedicated to the integration of complementary spatial information in hyperspectral data, decoding high-frequency details while preserving spectral characteristics. The blur estimation module focuses on estimating the blur kernel, thereby improving the sharpness of reconstructed results. To achieve this, a "one-to-one" brightness loss and a "one-to-multiple" high-frequency loss are formulated to constrain the learning of the fusion module, ensuring effective spatial-spectral information reconstruction. Additionally, an independent "one-to-one" blur brightness loss is introduced for the blur estimation module to decouple its learning from the fusion module. During training, the network takes blurred downsampled hyperspectral data as input, while the training target for the fusion module is the nonblurred downsampled data, and the training target for the blur estimation module is the blurred downsampled data.

The contributions of this work are as follows.

1) A novel computational HSI technique—aperture array-based self-supervised hyperspectral SR imaging approach—is proposed, enabling a compact imaging system with high spatiotemporal resolution.
2) A self-supervised hyperspectral SR model is introduced. A physics-guided attention-driven dual-stage network architecture is designed, combined with a divide-and-conquer self-supervised learning strategy. This alleviates the reliance on large-scale HR hyperspectral datasets required by existing methods.
3) Generalization analysis and real-world validation—Extensive simulations are conducted to evaluate the generalization of the proposed method to unknown degradations. Furthermore, a prototype system is constructed and validated on real-world hyperspectral data.

II. Related Works

A. Auxiliary Hyperspectral Image Super Resolution

The auxiliary hyperspectral SR approaches, which rely on HR color image, panchromatic image, or MSI as auxiliary information to be fused with low-resolution (LR) HSI to enhance spatial resolution. Kawakami et al. [20] employed nonnegative matrix factorization methods to reconstruct HR-HSI from paired LR-HSI and HR-MSI. Dian et al. [29] introduced nonlocal sparse tensor decomposition to preserve the 3-D spatial–spectral structure of HSI data. Wang et al. [30] introduced the haze-line prior and proposed a haze correction-based low-rank tensor completion method for pansharpening to acquire HR-MSI. Inspired by the success of deep learning in solving inverse problems and image restoration, deep neural networks have also been applied to hyperspectral SR. Initially, deep learning-based methods followed a purely data-driven approach. Lu et al. [31] borrowed the concept of transfer learning, where the mapping learned from LR images to HR images was transferred to the LR-HSI to HR-HSI mapping. To ensure spectral fidelity in reconstruction, they further applied collaborative nonnegative matrix factorization to the network output. Several network architectures have been designed to enhance image fusion capabilities, including 3-D convolutional networks [21], multiscale structures [32], and attention mechanisms [33]. However, these supervised learning approaches require large-scale training datasets with paired HR ground-truth images, which are challenging to obtain in real-world applications. To address this limitation, Qu et al. [22] proposed an unsupervised encoder–decoder architecture consisting of two encoder-decoder networks with shared decoder parameters, where sparse Dirichlet distribution constraints were imposed on the outputs of both decoders. Subsequently, Liu et al. [34] introduced a model-based unsupervised HSR

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network that integrates the nonnegative matrix factorization of HR-HSI into an autoencoder framework. Cao et al. [35] designed an unsupervised joint training framework for the three subnetworks, which can recover the HR-HSI without preknown degradation information. In this work, the HSI system captures only LR-HSI, making auxiliary hyperspectral image SR networks nondirectly applicable to this system. However, their fusion modules and unsupervised learning strategies can provide useful inspiration for HSR of this system.

B. Single Hyperspectral Image Super Resolution

Single hyperspectral image super resolution does not require auxiliary information and relies solely on LR-HSI, making it more practical for real-world applications. Consequently, it has attracted increasing attention. By leveraging the spectral correlations and the complementary spatial information between bands, hyperspectral SR enhances spatial resolution. However, due to the high dimensionality of hyperspectral data and distinct spectral patterns, single hyperspectral image SR remains a challenging problem. Early single hyperspectral image SR methods formulated HR-HSI reconstruction as an optimization problem by incorporating data fidelity terms and handcrafted priors [23], [24], such as self-similarity, sparsity, or low-rank constraints. However, these handcrafted priors are computationally expensive and struggle to accurately model the inherent spatial–spectral properties of hyperspectral images. Benefiting from the strong feature representation capabilities of deep neural networks, deep learning based methods have gradually been applied to single hyperspectral image SR. Given the high similarity among adjacent spectral bands, several two-stage or recurrent network structures have been designed for this task. Liu et al. [25] proposed a spectral grouping and attention-based residual dense network, where intergroup and intra-group feature extraction is performed in the early stages, and a spectral attention mechanism is applied to recalibrate features, ensuring spatial enhancement and spectral consistency. In deeper network layers, a residual dense spectral attention module aggregates hierarchical features. Later, Li et al. [4] introduced a dual-stage network architecture. In the coarse stage, a band-by-band SR approach is applied by utilizing information from adjacent bands. In the fine stage, an enhanced back-projection method with a spectral angle constraint is developed to ensure spatial–spectral consistency. To fully exploit the complementary and consecutive information among spectral bands, Wang et al. [26] proposed a recurrent feedback network. Zhang et al. [5] designed an improved Transformer-based single-image hyperspectral SR network, where SCC-kernel-based self-attention replaces the original attention matrix to better capture hyperspectral dependencies. Huang et al. [36] proposed the spectral clustering SR reconstruction algorithm based on a multispectral camera array. When synthesizing the multispectral data cube, this algorithm requires a trade-off between spatial information preservation and spectral fidelity, limiting its reconstruction performance for spectral data with large spectral spans. Moreover, it relies on a traditional Bayesian iterative algorithm, which necessitates a compromise between computational efficiency and reconstruction time. Recently, Pang et al. [6] adopted diffusion models for unsupervised hyperspectral SR, formulating the HR-HSI reconstruction as an estimation problem involving a reduced spectral-dimension image and its corresponding coefficient matrix. Miao et al. [37] proposed a self-supervised denoising diffusion spatiospectral model for hyperspectral image restoration, which combines deep image priors and diffusion models without requiring external training data. He et al. [38] introduced a spectral low-rank loss function for self-supervised learning, which was applied to hyperspectral image reconstruction from coded measurements.

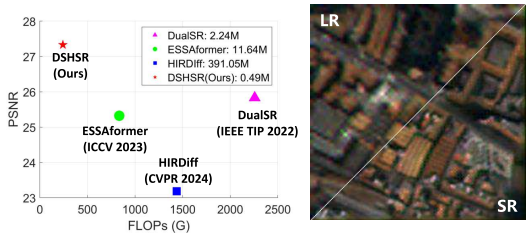


Fig. 1. PSNR versus FLOPs. Left: Comparison of our proposed method (DSHSR) with SOTA single hyperspectral image SR algorithms [4], [5], [6] on the PaviaU and Hyperspectral Project simulated remote sensing datasets. Numbers in legend denote network parameters. Right: Comparison between the observed hyperspectral image and the reconstructed hyperspectral image. The image size before reconstruction is 64 × 64, and after reconstruction, it is 256 × 256.

Existing single hyperspectral image SR methods predominantly adopt a supervised learning paradigm, requiring large-scale paired datasets of LR-HSI and HR-HSI, which is impractical given the inherent data scarcity in HSI. Moreover, supervised learning performs well in-distribution but is highly susceptible to distribution shifts. Recent works have employed diffusion models for unsupervised learning. However, diffusion models require multiple iterative optimizations, leading to significantly longer reconstruction time compared to feedforward models, and the number of iterations is challenging to determine adaptively. Some studies have pioneered the application of self-supervised learning in hyperspectral image denoising. However, no existing work has yet achieved self-supervised hyperspectral SR. While dual-stage network structures can enhance the fidelity of hyperspectral image reconstruction, current dual-stage architectures still cannot achieve end-to-end reconstruction.

III. Proposed Method

The proposed aperture array-based self-supervised hyperspectral SR imaging technique consists of a hardware encoding module and a software decoding module. The hardware encoder adopts a multiaperture array structure that is more conducive to practical engineering implementation, as shown in Fig. 2(a). A narrowband filter array is used to capture the hyperspectral information of the scene, which is then encoded onto the detector's imaging plane using a compressed sampling strategy at a frequency below the diffraction limit. The observed hyperspectral data is decoded through computational reconstruction algorithms to obtain clear, high signal-to-noise ratio (SNR), and high spatial resolution hyperspectral images. Fig. 2(c) illustrates the theoretical basis for applying

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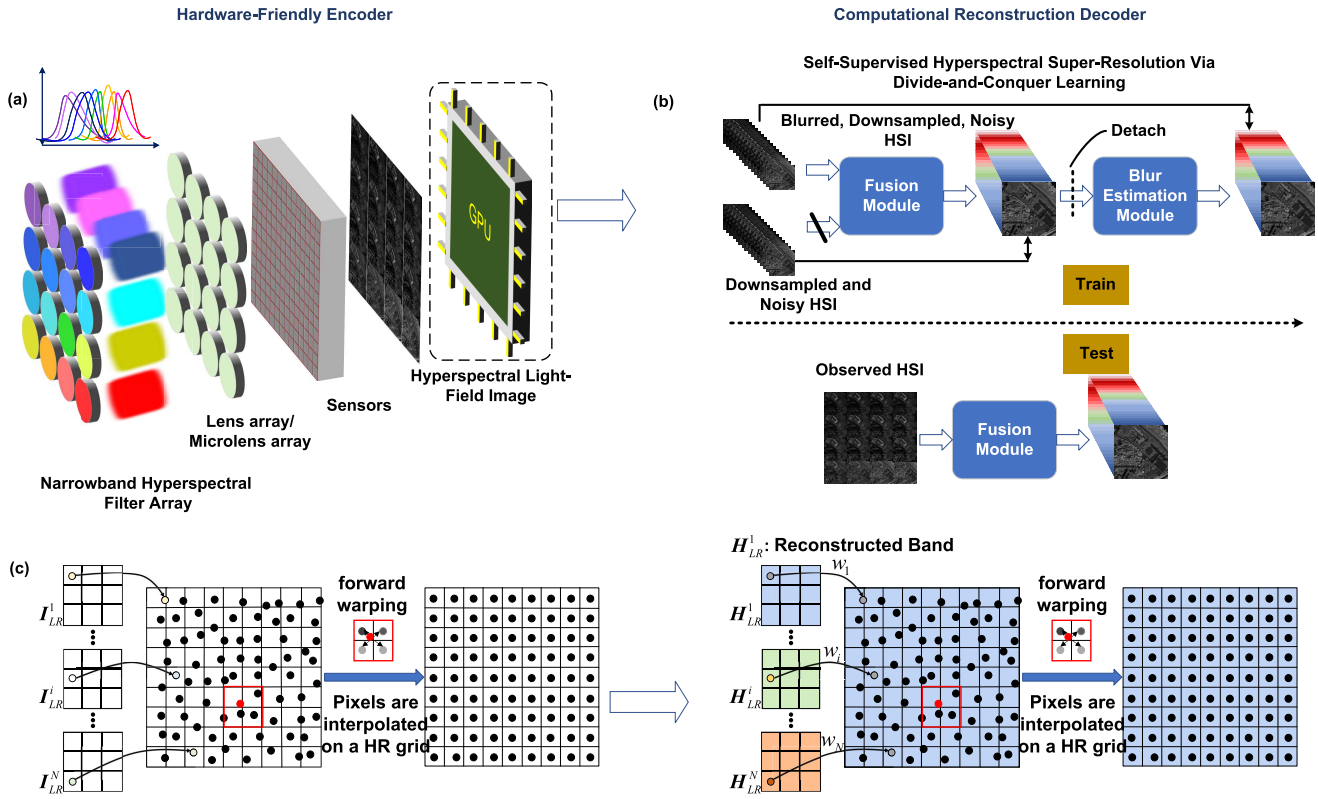


Fig. 2. Working principle of aperture array-based self-supervised hyperspectral SR imaging. (a) and (b) Illustrate this computational HSI paradigm, where LR hyperspectral data are rapidly acquired at the hardware level, and HR hyperspectral data are reconstructed through inverse optimization at the software level. (b) Presents the proposed divide-and-conquer self-supervised hyperspectral SR framework. (c) Demonstrates the basis for applying self-supervised learning to hyperspectral SR imaging based on an aperture array, where different apertures project onto different locations within a common coordinate system. By incorporating an attention mechanism, the concept of self-supervised monochromatic multiaperture imaging is extended to HSI.

self-supervised learning in hyperspectral SR imaging using an aperture array, where pixels from different apertures are mapped to different spatial locations in a common coordinate. Based on this principle, a divide-and-conquer self-supervised hyperspectral SR framework is designed, as shown in Fig. 2(b).

A. Imaging Model

Due to the high-dimensional nature of hyperspectral data, the physical limitations of hardware, and data transmission saturation, snapshot HSI systems face significant challenges. The scene information transmitted through the optical system is compressed and encoded in the discretized, quantized sensor. As a result, the captured hyperspectral data suffer from insufficient spatial resolution, hindering the precise capture of spectral information. Importantly, the rich scene information is not lost but is instead encoded in an aliased manner within the observed hyperspectral data. In this work, the forward imaging model for a snapshot hyperspectral SR imaging system based on an aperture array is analyzed. The system is a passive imaging system that uses ambient light or target-emitted radiation instead of active light sources. Consequently, the effects caused by scattering are minimized. The observation model of the system can be expressed as follows [36]:

$$\boldsymbol{I}^n_{\mathrm{BLR}} = \boldsymbol{D}\boldsymbol{B}_n\boldsymbol{W}_n\boldsymbol{S}_n\boldsymbol{I}_{\mathrm{HR}} + \boldsymbol{e}_n \quad (n = 1, 2 \ldots N). \tag{1}$$

In the equation, $\boldsymbol{I}^n_{\mathrm{BLR}}$ represents the image captured by the nth aperture in the aperture array; $\boldsymbol{I}_{\mathrm{HR}}$ represents the HR panchromatic image at the viewpoint of reference aperture; $\boldsymbol{W}_n$ is the geometric warping matrix for the nth aperture image relative to the reference aperture image, and $\boldsymbol{e}_n$ accounts for noise. N indicates the total number of apertures. $\boldsymbol{B}_n$ corresponds to the aberration blur and defocus blur of the optical system. In this work, $\boldsymbol{B}_n$ does not account for nonlinear distortions caused by atmospheric turbulence, and is therefore assumed to be shift-invariant. $\boldsymbol{D}$ represents 2-D downsampling operation on the imaging focal plane array, which includes blur due to photon integration on the sensor. $\boldsymbol{e}_n$ denotes the noise, whose distribution depends on the imaging environment. Under normal lighting conditions, image noise is primarily caused by detector readout noise, which is commonly modeled as Gaussian-distributed. In low-light environments, image noise also includes shot noise, which follows a Poisson distribution. $\boldsymbol{S}_n$ is the optical filtering matrix, a parameter related to the filter bandwidth and spectral characteristics of the captured scene. Let $\boldsymbol{S}_n\boldsymbol{I}_{\mathrm{HR}}$ be denoted as $\boldsymbol{I}^n_{\mathrm{HR}}$. $\boldsymbol{I}^n_{\mathrm{HR}}$ denote the HR image of the nth spectral band to be estimated. $(\boldsymbol{I}^n_{\mathrm{BLR}})_{n=1}^N$ and $(\boldsymbol{I}^n_{\mathrm{HR}})_{n=1}^N$ are referred to as $\boldsymbol{H}_{\mathrm{BLR}} \in \mathbb{R}^{h\times w\times N}$ and $\boldsymbol{H}_{\mathrm{HR}} \in \mathbb{R}^{(\lfloor\cdot\rfloor)^{1/2}Nh\times(\lfloor\cdot\rfloor)^{1/2}Nw\times N}$, respectively. Therefore, the observation model of the system can also be expressed as [36]

$$\boldsymbol{H}^n_{\mathrm{BLR}} = \boldsymbol{D}\boldsymbol{B}_n\boldsymbol{W}_n\boldsymbol{H}^n_{\mathrm{HR}} + \boldsymbol{e}_n \quad (n = 1, 2 \ldots N). \tag{2}$$

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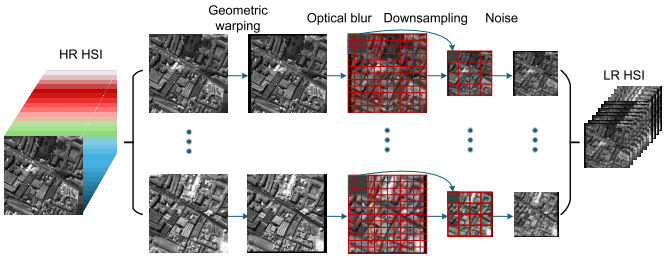


Fig. 3. Forward process of the aperture array-based hyperspectral SR imaging system.

Compared to circular layouts, the proposed aperture array system adopts a rectangular grid array to improve spatial efficiency and construct a more compact system architecture. This design maximizes the overlap of the fields of view between apertures, thereby supporting the largest possible super-resolved field of view, while also reducing alignment precision requirements. Since this work focuses on spatial SR, the narrowband filters used in different apertures are designed with nonoverlapping FWHM. This configuration allows the system to capture more diverse and complementary spectral features.

B. Observed Hyperspectral Data Characteristics

The data acquired by the multiaperture array HSI system exhibits the following characteristics.

1) Parallax between spectral images from different apertures. The spectral images captured by different apertures exhibit subpixel misalignment, as shown in Fig. 3(e).
2) Significant brightness differences between spectrally distant bands. As illustrated in Fig. 3(a). This is due to the wavelength-dependent response of objects, which results in different reflection or radiation intensities.
3) Texture structure differences between spectrally distant bands. As depicted in Fig. 3(b), variations in texture structures can occur between spectral bands, including reversed or missing textures in certain bands.
4) High similarity between adjacent spectral bands. As shown in Fig. 3(d), the left presents the correlation coefficients between each spectral band and its adjacent band across 32 scenes in the CAVE dataset [39], with all coefficients exceeding 0.955. The right presents correlation coefficients from the PaviaU dataset, all of which are greater than 0.9.
5) Noise, blur, and aliasing artifacts in individual band images. Aliasing artifacts, as shown in Fig. 3(c), arise because the sensor's sampling frequency is insufficient to fully restore the scene information passing through the optical system.

C. Attention-Driven Dual-Stage HSR Network Architecture

Based on the statistical characteristics of the observed data, the designed hyperspectral SR network must autonomously recognize regions with different texture structures to avoid introducing erroneous details while ensuring spectral brightness fidelity. Additionally, it must fully exploit the complementary spatial information of adjacent spectral bands to eliminate aliasing artifacts and recover high-frequency details. Furthermore, the network should enhance fine details, improve sharpness, and suppress noise.

To achieve these objectives, the proposed DSHSR network consists of a fusion module and a blur estimation module, as illustrated in Fig. 4. The proposed fusion module adopts an attention-driven dual-stage network architecture. Specifically, the coarse fusion module incorporates an attention mechanism to adaptively integrate complementary spatial information from adjacent spectral bands, enabling high-frequency detail recovery. The fine fusion module employs a progressive dilated 3-D convolutional network structure to expand the receptive field along the spectral dimension, ensuring spectral fidelity in the reconstructed output.

As illustrated in Fig. 4, both blurred and nonblurred downsampled hyperspectral images are used as training targets. The blurred downsampled hyperspectral image is fed into the hyperspectral SR network. Due to variations in brightness and texture structures across spectral bands, the optical flow estimation networks designed for monochromatic images perform poorly on spectral data. To address this issue, the Structure Consistency Boosting transform [40] is applied before optical flow estimation to enhance structural similarity across spectral bands. Subsequently, optical flow estimation is performed on the transformed spectral data. The FNet [41] is adopted to compute the offset between the reference aperture image and other aperture images, which is denoted as "ME" in Fig. 4. Based on the calculated offsets, the subpixel motion compensation (SPMC) [42] that is a forward-warping (FW) alignment strategy is adopted to align the parallax between apertures, preserving subpixel complementary spatial information. The aligned features are then fused band by band through coarse fusion. Finally, fine fusion is applied to the coarsely fused hyperspectral features to reconstruct the spectral data and generate the output. The blur estimation network [43] is used to estimate the blur kernel of the input. The estimated blur kernel is applied to the output SR hyperspectral image from the fusion module to generate a blurred SR hyperspectral image. During inference, the output of the fusion module serves as the final result.

1) Luminance-Texture Separation: As shown in Fig. 3, there exist brightness and texture differences among images of different spectral bands. Since objects with different colors and materials exhibit distinct spectral characteristics, the global brightness or texture differences are nonlinear. Due to this brightness difference, fusing features from different spectral bands to enhance spatial resolution while ensuring spectral fidelity is challenging. Meanwhile, the goal of hyperspectral SR in this work is to improve the clarity of texture details and restore high-frequency information. Therefore, a separation of luminance and texture components is performed. The LR blurred hyperspectral high-frequency image $\boldsymbol{HH}_{\mathrm{BLR}}$ is fed into the SR network, generating a HR hyperspectral high-frequency image as output. Simultaneously, the low-frequency hyperspectral image is upsampled to match the super-resolved image

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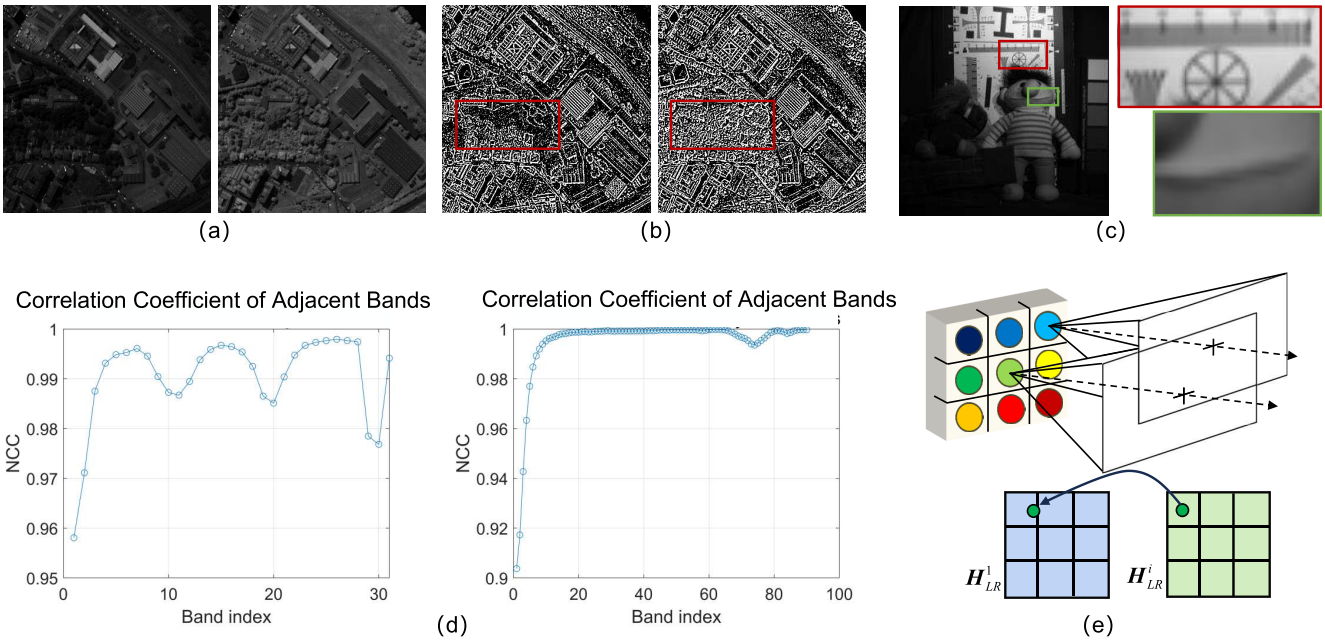


Fig. 4. Statistical characteristics of observed data in the multiaperture array hyperspectral SR imaging system. (a) Brightness differences. (b) Texture structure differences. (c) Aliasing high-frequency artifacts. (d) Similarity between adjacent spectral bands. (e) Parallax and subpixel offset between apertures.

size. A three-layer fully convolutional network is designed to sharpen the upsampled low-frequency hyperspectral image. As shown in Fig. 4, the sharpened result is then added to the HR hyperspectral high-frequency image output by the fusion module, resulting in the final HR hyperspectral data cube. Additionally, the separation of luminance and texture components also facilitates network denoising. Since noise is the high-frequency component, it is contained within the texture component. Through the coarse fusion module, noise in the texture component can be effectively removed

$$\boldsymbol{LH}^{s}_{\mathrm{BLR}} = \mathrm{SL}\left(BU\left(\boldsymbol{LH}_{\mathrm{BLR}}\right), \theta_{\mathrm{SL}}\right) \in \mathbb{R}^{h\times w\times C}. \tag{3}$$

In the equation, BU(·) represents the bilinear upsampling operation, SL(·) denotes a three-layer fully convolutional network designed to sharpen the upsampled low-frequency hyperspectral image. $\boldsymbol{LH}_{\mathrm{BLR}}$ represents the low-frequency component of the input $\boldsymbol{H}_{\mathrm{BLR}}$.

2) Coarse Fusion: Texture-Aware Attention Fusion: During the coarse fusion stage, SR reconstruction is performed band by band to generate detail-enhanced super-resolved hyperspectral features. As shown in Fig. 5(a), the coarse fusion process for the nth spectral band as the target band is illustrated. Ten neighboring spectral bands around the nth band are selected, resulting in a total of 11 bands for feature extraction. The feature extraction module consists of a convolutional network with two stacked residual blocks. To achieve effective spatial information fusion of high-frequency features from different spectral bands with varying texture structures, the texture-aware attention coarse fusion module is designed. First, the 10 spectral band features excluding the target band are aligned using SPMC [42], utilizing optical flow estimated by the ME module, resulting in the upsampled and aligned features $\boldsymbol{X}^{n}_{\mathrm{HRA}} \in \mathbb{R}^{([)^{1/2}]Nh\times([)^{1/2}]Nw\times C}$. Simultaneously, FW alignment is applied to these 11 spectral band features, producing another set of aligned features $\boldsymbol{X}^{n}_{\mathrm{LRA}} \in \mathbb{R}^{h\times w\times C}$. As shown in Fig. 5(b), the weight predictor is designed to perceive feature differences across spectral bands. $\boldsymbol{X}^{n}_{\mathrm{LRA}} \in \mathbb{R}^{h\times w\times C}$ is fed into the module, which outputs the fusion weights

$$\boldsymbol{A}^{n+i} = WP\left(\boldsymbol{X}^{n}_{\mathrm{LRA}}, \boldsymbol{X}^{n+i}_{\mathrm{LRA}} - \boldsymbol{X}^{n}_{\mathrm{LRA}}, \theta_{WP}\right) \in \mathbb{R}^{h\times w\times C}. \tag{4}$$

In the equation, WP(·) represents the weight predictor. Specifically, $\boldsymbol{X}^{n}_{\mathrm{LRA}}$ and $\boldsymbol{X}^{n+i}_{\mathrm{LRA}}$ are passed through the convolutional layer to reduce the channel dimension. The difference between the target band feature with reduced channel dimensions and the feature of the bands to be fused, along with the target band feature itself, is concatenated and fed into the residual layer to generate the fusion weight. The obtained weights for different bands are then normalized using the ReLU function, producing the final normalized weights. The attention-weighted features from all bands are summed to obtain the coarse fusion result for the nth band, as shown in the following equation:

$$\boldsymbol{X}^{n}_{\mathrm{CF}} = \sum_{i-5}^{i+5}\left(BU\left(\boldsymbol{A}^{n+i}\right)\odot \boldsymbol{X}^{n+i}_{\mathrm{HRA}}\right) \in \mathbb{R}^{\sqrt{N}h\times\sqrt{N}w\times C}. \tag{5}$$

During training, i is not equal to 0.

3) Fine Fusion: Progressive 3-D Spectral Reconstruction: The fine fusion stage adopts a progressive dilated 3-D convolutional network. By leveraging 3-D convolutions, the network captures the correlations among different spectral bands to ensure high-fidelity spectral reconstruction. Thanks to the spatial information interaction achieved in the coarse fusion stage, the fine fusion process requires fewer parameters while still ensuring high-fidelity reconstruction. Consequently, the fine fusion module consists of only three 3-D convolutional

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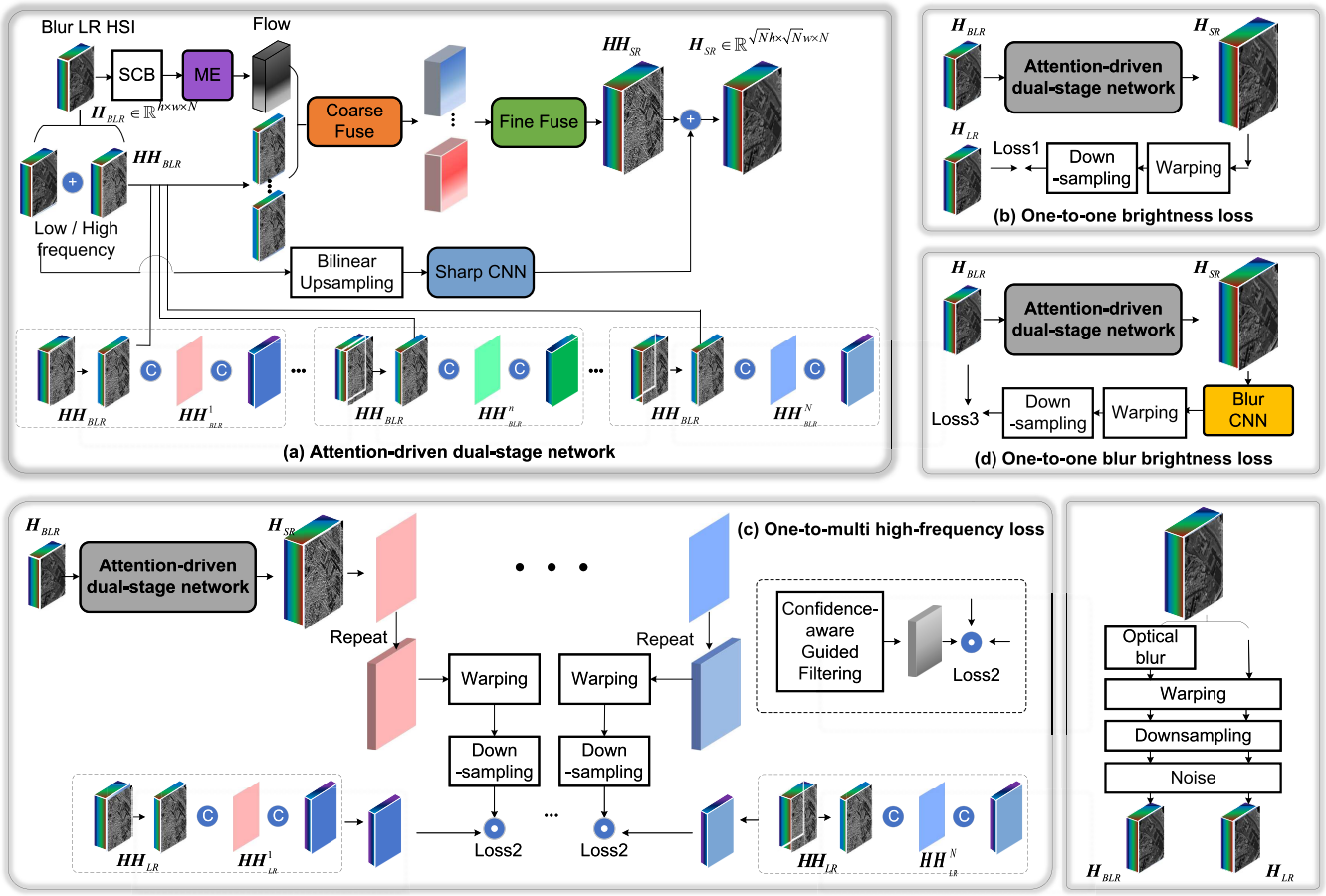


Fig. 5. Attention-driven dual-stage network architecture and divide-and-conquer self-supervised learning strategy flowchart. (a) Blurred LR-HSI is fed into the attention-driven dual-stage network. (b) Nonblurred LR-HSI $\boldsymbol{H}_{\mathrm{LR}}$ is used as the training target to constrain the learning of the attention-driven dual-stage network. (c) High-frequency components of nonblurred LR-HSI $\boldsymbol{HH}_{\mathrm{LR}}$ serve as training targets to constrain the attention-driven dual-stage network, focusing on high-frequency information learning. (d) Input blurred LR-HSI is used as the training target to constrain the learning of the blur estimation network.

layers, as shown in Fig. 7. To effectively learn spectral features across a wide range, increasing the number of convolutional channels along the spectral dimension is necessary. However, this leads to a significant increase in both model parameters and computational complexity. As shown in Fig. 3, adjacent hyperspectral bands exhibit strong similarity. Therefore, we employ dilated 3-D convolutions to expand the receptive field along the spectral dimension without increasing the number of parameters or computational complexity

$$\boldsymbol{HH}_{\mathrm{SR}} = FF\left(\left\{\boldsymbol{X}_{\mathrm{CF}}^{n}\right\}_{n=1}^{N}, \theta_{FF}\right) \in \mathbb{R}^{\sqrt{N}h\times\sqrt{N}w\times N}. \tag{6}$$

$$\boldsymbol{H}_{\mathrm{SR}} = \boldsymbol{LH}_{\mathrm{BLR}}^{s} + \boldsymbol{HH}_{\mathrm{SR}}. \tag{7}$$

D. Divide-and-Conquer Self-Supervised Hyperspectral Learning

To achieve simultaneous detail restoration, denoising, and deblurring, the divide-and-conquer self-supervised learning strategy is designed. Specifically, three loss functions are designed for task-specific learning in the network.

1) The one-to-one brightness loss minimizes the difference between the geometrically transformed and downsampled output of the attention-driven dual-stage network and the LR HSI. The one-to-one brightness loss ensures that each spectral band is supervised solely by its corresponding LR band and maintains luminance fidelity.
2) The one-to-multi high-frequency loss enhances reconstruction quality using multiple LR bands to jointly supervise the reconstruction of a single band, thereby promoting high-frequency detail recovery in the attention-driven dual-stage network. The participating fusion bands and training target band are nonoverlapping to prevent noise equivalency mapping, ensuring simultaneous denoising during the high-frequency detail fusion process.
3) The one-to-one blur brightness loss constrains the geometrically transformed and downsampled output of the blur estimation network to match the blurred LR HSI, ensuring that the attention-driven dual-stage network produces sharp and clean reconstructions.

1) One-to-One Brightness Loss: Since the parameters of the model are independent of the number of spectral bands, different numbers of bands can be used for fusion during training and testing. Taking the nth band as the target band for spatial information fusion as an example. During training, the high-frequency features of 10 neighboring bands around the nth band are used for fusion, while the nth band itself is

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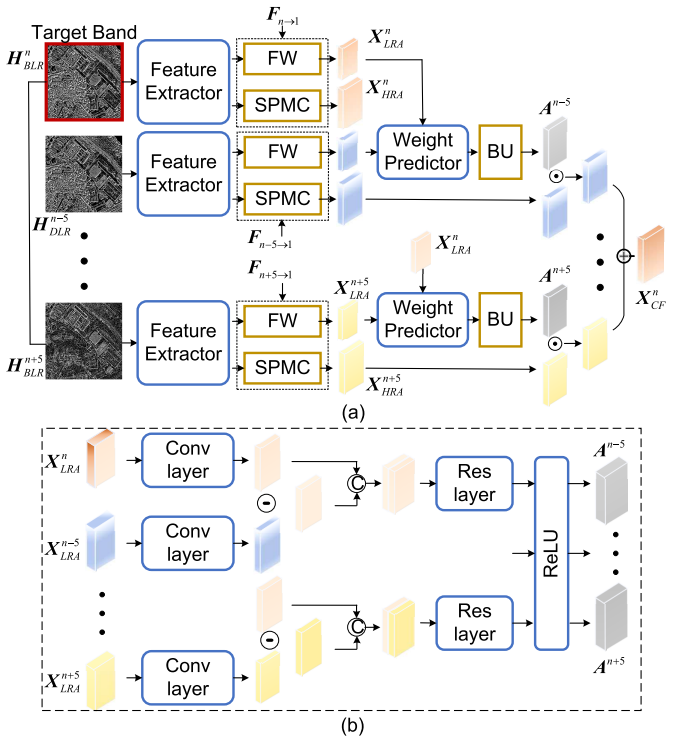


Fig. 6. Coarse Fusion Module and Weight Predictor Flowchart. ME represents the Motion Estimation module. SPMC denotes SPMC. FW stands for Forward Warping Alignment. BU refers to Bilinear Upsampling. (a) Coarse fusion: texture-aware attention fusion. (b) Weight predictor.

only used for computing the fusion weights. During testing, the high-frequency features of all 11 bands, including the nth band, are used for fusion. The nonblurred LR image of the nth band is used to supervise the luminance reconstruction of the nth band image. By adopting a blind-spot learning strategy that excludes the target band, the predicted values from mapping noise patterns identical to those in the training targets

$$\mathcal{L}_1 = \frac{1}{N}\sum_{n=1}^{N}\left\| \boldsymbol{H}^n_{\mathrm{LR}} - D\left(W\left(\boldsymbol{H}^n_{\mathrm{SR}}, \sqrt{N}\boldsymbol{F}_{n\to 1}\right)\right)\right\|_1 \tag{8}$$

where $W()$ represents the backward warping operation based on the flow and $D()$ represents the downsampling operation.

2) One-to-Multi High-Frequency Loss: To constrain the SR reconstruction of the high-frequency image of the nth band using LR high-frequency images from other bands, we design the one-to-multi high-frequency loss. Specifically, the 10 LR nonblurred high-frequency band images closest to the nth band, excluding the 10 bands input in the coarse fusion stage, are selected as supervision targets

$$\mathcal{L}_2 = \frac{1}{10\cdot N}\sum_{n=1}^{N}\sum_{i=1}^{10}\left\| \boldsymbol{A}_{GF}^{(m(n,i),n)} \odot \left(\boldsymbol{H}\boldsymbol{H}^{m(n,i)}_{\mathrm{LR}} - D\left(W\left(\boldsymbol{H}\boldsymbol{H}^n_{\mathrm{SR}}, \sqrt{N}\boldsymbol{F}_{m(n,i)\to 1}\right)\right)\right)\right\|_1. \tag{9}$$

In the equation, $m()$ represents the selection of the 10 band indices closest to the nth band from the remaining bands. $\boldsymbol{A}_{GF}$ denotes the explicit attention map computed using the guided filtering residuals [40] to distinguish exclusive texture regions between the two bands.

3) One-to-One Blur Brightness Loss: The estimated blur kernel from the blur estimation network is convolved with the output of the fusion module to produce $\boldsymbol{H}_{\mathrm{BSR}}$. To train the blur estimation network, the LR blurred hyperspectral image $\boldsymbol{H}_{\mathrm{BLR}}$ is used as the supervision signal, as described in (9), by minimizing the difference between the downsampled image of $\boldsymbol{H}_{\mathrm{BSR}}$ and $\boldsymbol{H}_{\mathrm{BLR}}$. The function torch.detach is employed to detach $\boldsymbol{H}_{\mathrm{BSR}}$ and $\boldsymbol{H}_{\mathrm{SR}}$ from the computational graph during backpropagation, ensuring that $\boldsymbol{H}_{\mathrm{BLR}}$ only constrains the learning of the blur estimation network. This approach guarantees that the output of the fusion module remains sharp

$$\mathcal{L}_3 = \frac{1}{N}\sum_{n=1}^{N}\left\| \boldsymbol{H}^n_{\mathrm{BLR}} - D\left(W\left(\boldsymbol{H}^n_{\mathrm{BSR}}, \sqrt{N}\boldsymbol{F}_{n\to 1}\right)\right)\right\|_1. \tag{10}$$

The overall loss function is as follows:

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_1 + \lambda\mathcal{L}_2 + \mathcal{L}_3 + \mu\mathcal{L}_k \tag{11}$$

where $\mathcal{L}_k$ is the blur kernel sparsity loss from [43], λ is 50, μ is 0.04.

IV. Experiments

A. Dataset and Implementation Details

For the simulated HSI training dataset, publicly available datasets are used as the ground-truth hyperspectral images, including the Harvard dataset [44] and Chikusei dataset [45] with remote sensing images, the CAVE partial dataset [39] and Time Lapse dataset [46] with natural images, and the Hyperspectral City dataset [47]. The ground-truth hyperspectral images are normalized and undergo a degradation process, including random geometric warping with shifts in the range of [−4,4], random Gaussian blur with a standard deviation (Std) randomly sampled from 0.4 to 2.0 [43], where the Gaussian kernel size is set to $int(6+1)$, 4× bilinear downsampling, and additive white Gaussian noise (AWGN) with a Std of 0.01 [48]. We first utilize the Adam optimizer and the cosine annealing scheduler to train FNet for 50 epochs. The trained FNet parameters are then imported into the motion estimation module of DSHSR. Next, the AdamW optimizer and the cosine annealing scheduler are used to train DSHSR for 100 epochs. During the training of DSHSR, the parameters of the motion estimation module remain unchanged. The cosine annealing strategy is deployed for steadily decreasing the learning rate from 10^{-4} to 10^{-6}.

For the simulated HSI testing dataset, publicly available datasets are also used, including Pavia Center and University, Indian Pines, Salinas, and Cuprite datasets with remote sensing images.,[1] Washington dc Mall dataset [49] with remote sensing images, and the CAVE partial dataset [39] with natural images. The test dataset contains scenes that do not overlap with those in the training dataset. The hyperspectral images undergo the same degradation process as the simulated training data. The Std of Gaussian blur for the test data is set to 1.0, and the AWGN Std is set to 0.01. Additionally, we test hyperspectral images with varying degrees of blur by setting the Std of the

[1]https://www.ehu.eus/ccwintco/index.php?title=Hyperspectral Remote Sensing Scenes.

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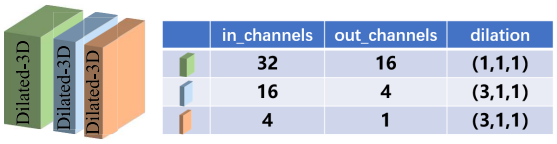


Fig. 7. Coarse fusion module: workflow and parameters.

TABLE I
QUANTITATIVE COMPARISON ON THE SIMULATED REMOTE SENSING DATASETS

	DualSR	ESSAformer	HIRDIff	DSHSR(Ours)
PSNR/dB↑	26.61	25.50	23.40	**28.50**
SSIM↑	0.73	0.66	0.47	**0.84**
SAM↓	11.13	14.14	16.04	**8.35**
Time/s↓	43.21	**0.08**	15.63	1.70
FLOPs/G↓	2257.00	834.16	1436.70	**241.60**
Para/M↓	2.24	11.64	391.05	**0.49**

Gaussian blur kernel within the range of [0.5,2.0] with an interval of 0.1, resulting in a total of 16 test sets. The noise Std for each set is fixed at 0.01.

To evaluate the generalization capability of the trained model in the out-of-distribution domain, test data are synthesized using different degradation parameters from those used in training. The Std of AWGN is set to [0.02, 0.03, 0.04, 0.05] to test the generalization capability of the trained model under more complex noise interference. Furthermore, we have built a prototype system to validate the proposed method's generalization capability in real-world scenarios.

We compare our method with three SOTA single hyperspectral image SR methods: DualSR [4], ESSAformer [5], and HIRDiff [6]. Among them, DualSR is a non-end-to-end supervised method, ESSAformer is an end-to-end supervised method, and HIRDIff is an end-to-end unsupervised method. The single hyperspectral image SR comparison methods take spatially aligned LR hyperspectral images as input. Therefore, the simulated LR hyperspectral images used as input for these methods are generated by removing geometric warping from the degradation process described above, while all other steps remain unchanged.

B. Simulation Results

Since the degradation process of the test data is consistent with that described in the papers of the comparison methods, including Gaussian blur, 4× downsampling, and AWGN, the pretrained models provided by these papers are used for testing. Figs. 8 and 9 present the test results of different methods on the simulated remote sensing dataset. The visualization results include three narrowband images corresponding to red, green, and blue wavelengths, as well as a pseudo-color image synthesized from these three bands. Additionally, the difference images between the reconstructed images and the ground truth are also displayed.

As shown in Figs. 8 and 9, our method DSHSR effectively restores details, enhances image clarity, and suppresses noise compared to the LR observed hyperspectral images. Fig. 10 presents the spectral response curves of different methods on the simulated CAVE dataset. The spectral angle mapper (SAM) values between the reconstructed and ground-truth images are calculated, where our method DSHSR achieves the lowest SAM value, indicating that the reconstructed spectral response curves align more closely with the ground truth. The spectral response curves for three different color regions are also provided for visualization. Table I presents the quantitative results of different comparison methods on the Pavia Center and University, Indian Pines, Salinas, Cuprite, and Washington dc Mall remote sensing datasets. Our method, DSHSR, outperforms all other approaches in performance metrics, achieving a PSNR value that is 1.9 dB higher than the second-best method.

DualSR introduces excessive sharpening that leads to ringing artifacts, as indicated by the white arrows in Figs. 9 and 10, and retains aliasing artifacts from the input LR HSI, which significantly hampers the recovery of high-frequency details, as highlighted by the red arrows in Fig. 8. As clearly observed in the colorchecker and doll shown in Fig. 10, ESSAformer suffers from spectral distortion and color shifts, with a SAM value that is 9.0 higher than that of DSHSR. In addition, its reconstructed images still exhibit aliasing artifacts that are not effectively restored, as evidenced by the red arrows in shown in Fig. 8. As shown in Figs. 8–10, the reconstructed results of HIRDiff exhibit more severe noise than the input LR-HSI. HIRDiff adopts a diffusion-based model, where the forward process progressively adds noise, and the reverse process aims to recover high-frequency details. However, since the number of reverse iterations is predefined and cannot adapt to the degradation level of the input image, residual noise persists in the reconstructed results. As reported in Table I, the PSNR is the lowest among all compared methods.

Fig. 11 presents a comparison of PSNR for reconstructed results under different levels of Gaussian blur on the simulated remote sensing datasets. As observed, the mean PSNR of ESSAformer and HIRDIff remains relatively stable across varying blur levels. The mean PSNR of our proposed method, DSHSR, gradually decreases as the blur level increases. Meanwhile, DualSR exhibits an initial increase in mean PSNR with increasing blur, followed by a gradual decline, with a turning point at a Gaussian blur Std of 1.6. At this point, the performance gap between DSHSR and DualSR narrows. This phenomenon occurs because, under excessive blurring, the nonredundancy between different spectral bands is significantly reduced, leaving only limited complementary spatial information. As a result, the SR performance of DSHSR degrades for highly blurred images. In practical imaging systems, optical aberrations are already corrected by the lens, preventing severe blur distortions. This has been further validated in real results. In Fig. 12, the diameter of each circle represents the Std. It can be seen that the Std over multiple runs remains nearly the same for ESSAformer, HIRDiff, and our method DSHSR across varying blur levels. For DualSR, however, the Std slightly increases with the blur level.

Additionally, our method DSHSR features lower model parameters and computational complexity. The runtime of all methods is measured on a GV100 GPU for 4× SR with an input image size of 64 × 64. For our method DSHSR, the reported inference time refers to the processing time of the

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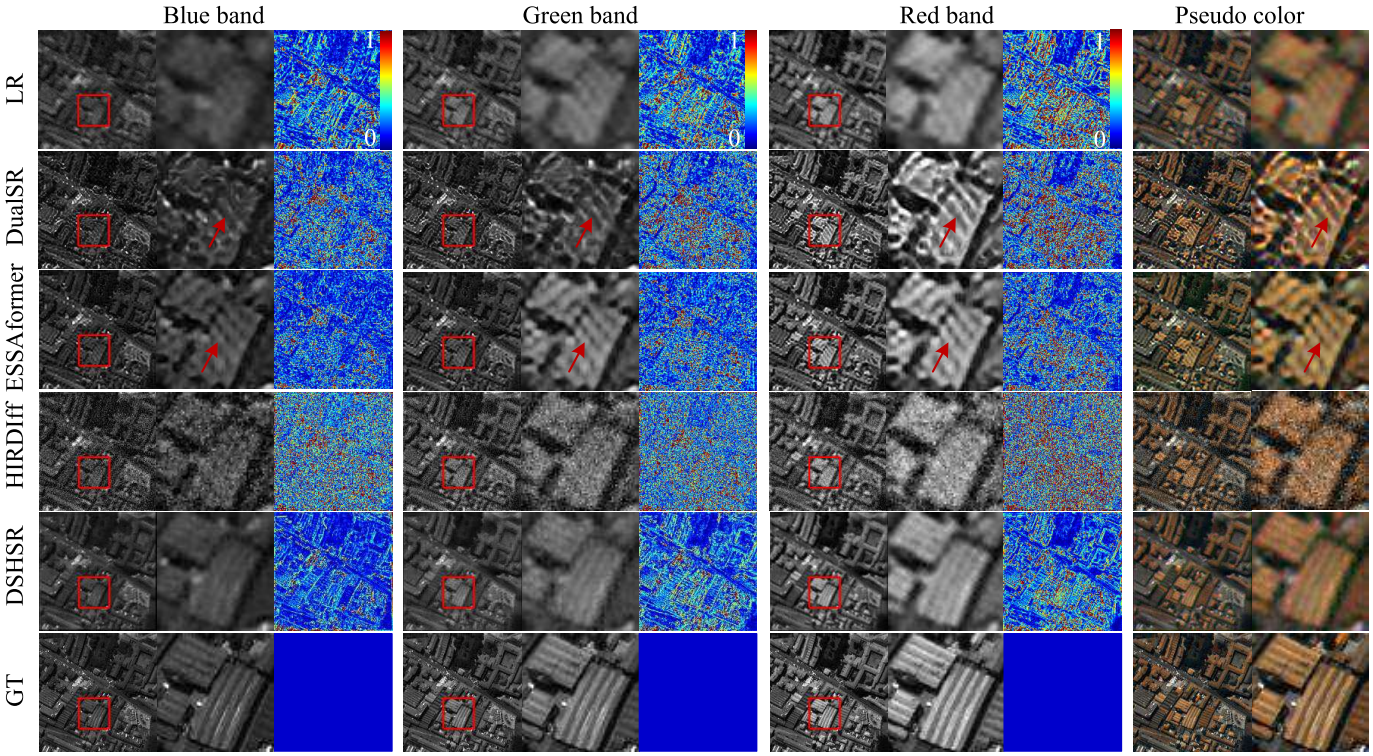


Fig. 8. Reconstruction comparison on the simulated PaviaU hyperspectral dataset[1]. DSHSR refers to the proposed method. For each band, the first column shows the reconstructed image, the second column presents the zoomed-in region highlighted by the red box, and the third column displays the difference between the reconstructed image and the ground truth. The selected bands include narrowband red, green, and blue, as well as a pseudo-color image synthesized from these three narrowband bands.

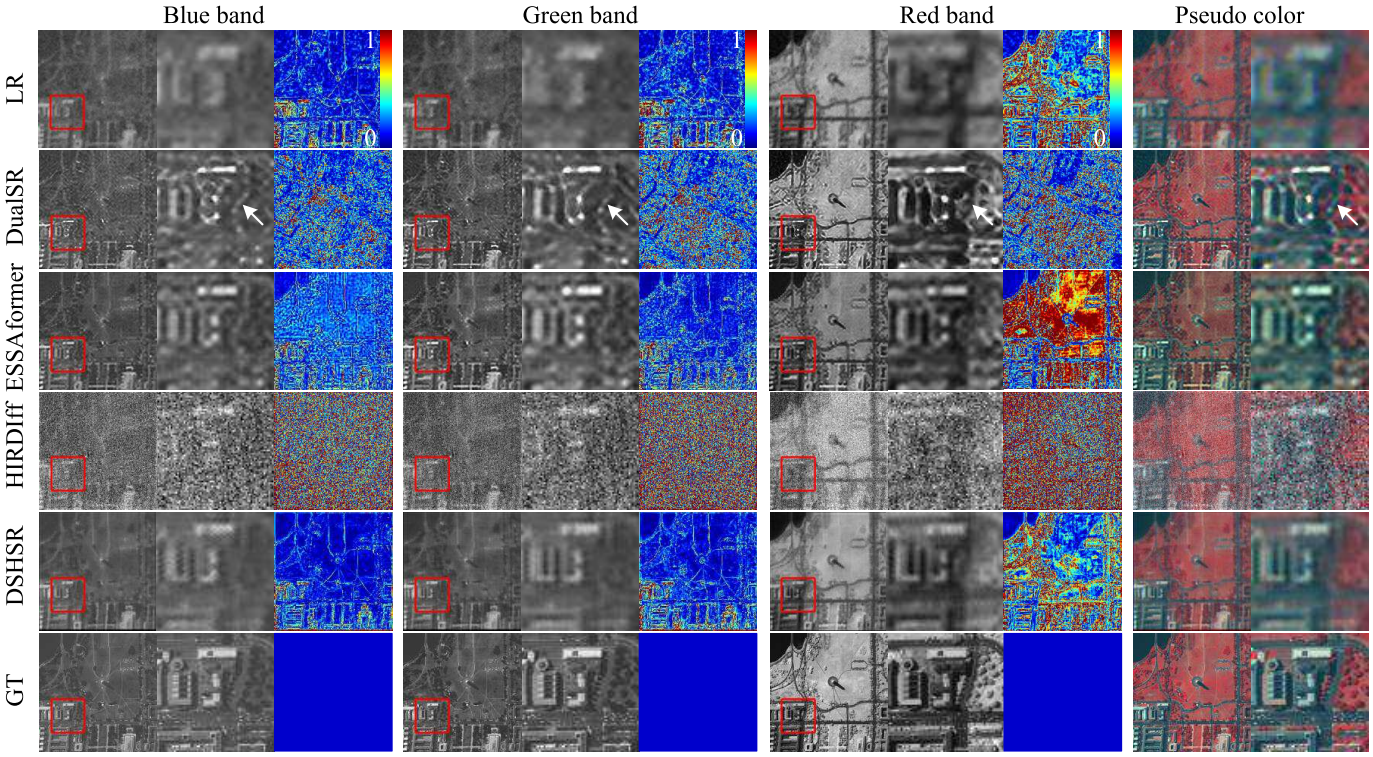


Fig. 9. Reconstruction comparison on the simulated Washington dc Mall dataset [49].

Attention-driven dual-stage network shown in Fig. 5(a)—from the input of the Blur LR HSI to the output of the HR-HSI from the fusion module. This runtime includes the time of the motion estimation module, but excludes the additional

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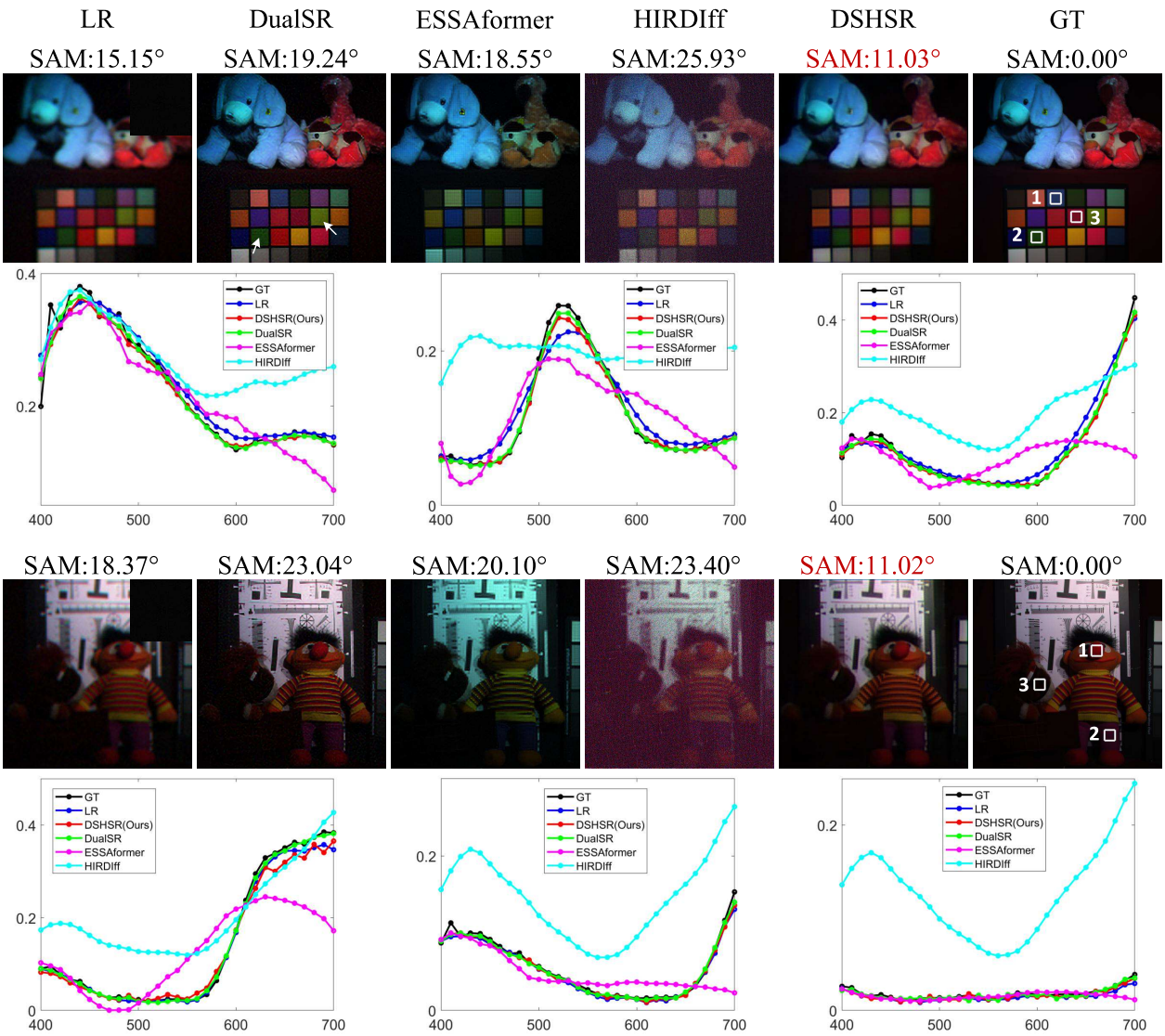


Fig. 10. Reconstruction spectral response comparison on the simulated CAVE dataset [39].

processing introduced by the self-supervised learning strategy shown in Fig. 5(b)–(d). DualSR adopts a dual-stage architecture, where the fine stage employs a back-projection method rather than a neural network and is implemented on the MATLAB platform. As a result, the reported runtime and FLOPs for DualSR do not include the fine stage. Even without it, DualSR already has the highest runtime and FLOPs among all methods, indicating that although it achieves the second-best reconstruction accuracy, it is the least efficient. Among the compared methods, ESSAformer achieves the fastest runtime. However, runtime depends not only on the network design but also on code optimization. While ESSAformer is 1.62 s faster than our DSHSI, its FLOPs are 592.56 G higher. HIRDiff is based on a diffusion model, where image generation is performed through an iterative optimization process. As a result, its runtime and FLOPs are 13.93 s and 1195.1 G higher than ours, respectively. These comparisons demonstrate that our DSHSI achieves a favorable trade-off between computational efficiency and reconstruction performance.

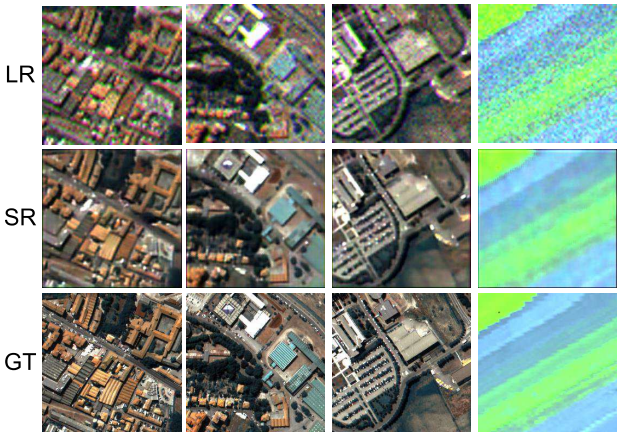


Fig. 11. Reconstruction results of our DSHSR under Poisson noise interference.

We also experiment with other types of synthetic noise. Poisson noise is the dominant source of noise in photographs.

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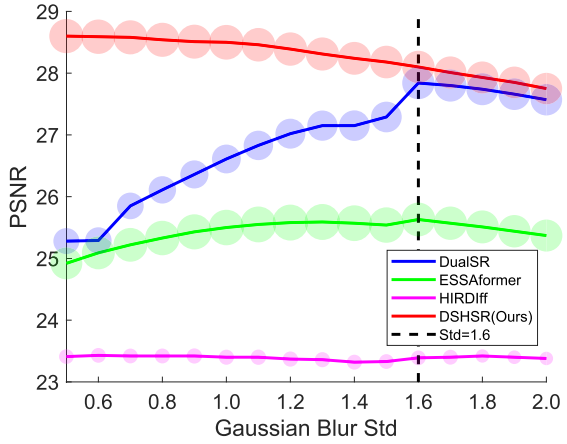


Fig. 12. Reconstruction comparison on the simulated remote sensing datasets under different levels of Gaussian blur. The diameter of the circles represents the Std.

TABLE II
GENERALIZATION ABILITY OF OUT-OF-DISTRIBUTION NOISE DOMAIN

PSNR/dB↑	DualSR	ESSAformer	HIRDIff	DSHSR(Ours)
0.02	25.17±2.05	25.32±3.25	23.41±1.45	**28.27±3.09**
0.03	23.67±1.54	25.03±3.00	23.38±1.32	**27.84±2.77**
0.04	22.28±1.22	24.67±2.76	23.38±1.28	**27.31±2.43**
0.05	21.03±1.02	24.22±2.50	23.33±1.26	**26.72±2.16**

In this experiment, Poisson noise is used instead of Gaussian noise, while all other settings remain the same to synthesize a degraded HSI for training the DSHSR network. The noise strength is controlled by γ, which is randomly sampled from the range [50, 500] during training. Testing is conducted on the previously described remote sensing datasets, with γ also randomly sampled from [50, 500]. The results before and after reconstruction are shown in Fig. 11. The PSNR of the HSI reconstructed by DSHSR is 28.03 dB, which is 3.22 dB higher than that of the observed LR-HSI.

C. Generalization Across Noise Levels

Generalization ability is mainly reflected in two aspects: different scenes and different degradation distributions. Figs. 8 and 9 demonstrate the generalization ability of our method in remote sensing scenes. With the rapid development of deep learning, generalization across different scenes has significantly improved, while generalization across different degradation distributions still requires further research. Fig. 13 illustrates the generalization ability of the trained model in the out-of-distribution domain by testing on simulated data with more severe noise interference than the training data, where the Std of the Gaussian blur kernel is 1.0. It can be observed that although the proposed method, DSHSR, is trained only on data with a noise variance of 0.01, the trained model effectively generalizes to data with more severe noise interference.

In Table II, the left value in each cell represents the mean PSNR, and the right value represents the Std. As shown in Fig. 13 and Table II, as the noise interference increases, the reconstruction performance of DualSR deteriorates. Compared to DualSR, ESSAformer, and HIRDIff exhibit lower sensitivity to noise, with no significant differences in reconstruction results under varying noise levels. However, ESSAformer suffers from spectral distortion and generates incorrect textures, while HIRDIff still retains residual noise in its reconstructions. In contrast, our method reconstructs clean images with high spectral fidelity, restoring fine texture details that faithfully match the ground truth. As shown in Table II, when the noise variance is 0.02, the proposed method DSHSR achieves a mean PSNR of 28.27. As the noise variance increases gradually by a step size of 0.01, the PSNR of DSHSR decreases steadily, with a trend of approximately 0.4, 0.5, and 0.6 dB. Compared to other methods, our method DSHSR consistently achieves the highest PSNR under different levels of noise interference. In comparison with DualSR, DSHSR also maintains more stable reconstruction performance.

D. Real Results

We conduct tests using real-world observed data collected by the multispectral camera array system built in [36]. The system prototype, component model specifications, and spectral response curves of the nine narrowband filters are shown in Fig. 14. In aperture array-based hyperspectral SR imaging, the parallelism of optical axes between apertures is a critical factor that affects both imaging performance and the feasibility of engineering applications. In a perfectly manufactured camera array system, significant redundancy may arise between apertures beyond a certain work distance, rendering SR ineffective [146]. However, in practical system fabrication and assembly, random deviations in optical axis parallelism are inevitable. These slight misalignments can actually introduce beneficial subpixel shifts between apertures, thereby increasing interaperture nonredundancy and enhancing SR capability. Compared to coded-aperture spectral imaging or diffractive optical spectral imaging, the proposed system does not require custom optical components. Instead, it can be built using widely available, mass-produced commercial spectral filters, lenses, and sensors, which significantly reduces hardware costs and improves the potential for real-world engineering implementation.

Fig. 15 shows a real scene captured using the constructed prototype system. The model trained on simulated data is directly applied to real-world data to verify that the proposed learning framework aims to guide the network in learning the fusion of information between spectral bands, rather than merely performing data-driven fitting. In a real-world scene, there exists a larger pixel shift between apertures. Therefore, the pixel shift range for the synthetic data is set to $[-16, 16]$, and FNet is retrained accordingly. The newly trained motion estimation module parameters are then used to replace those in the DSHSR model trained on simulated experiments. Our method allows for flexible adjustment of the motion estimation module based on actual requirements. The optical flow estimated by FNet is used to align the real spectral data with the camera array, and the aligned data are then fed into the comparison method networks.

Fig. 16 presents a comparison of reconstruction results on hyperspectral data collected by the multiaperture multispectral prototype system. As observed in Fig. 16, the hyperspectral

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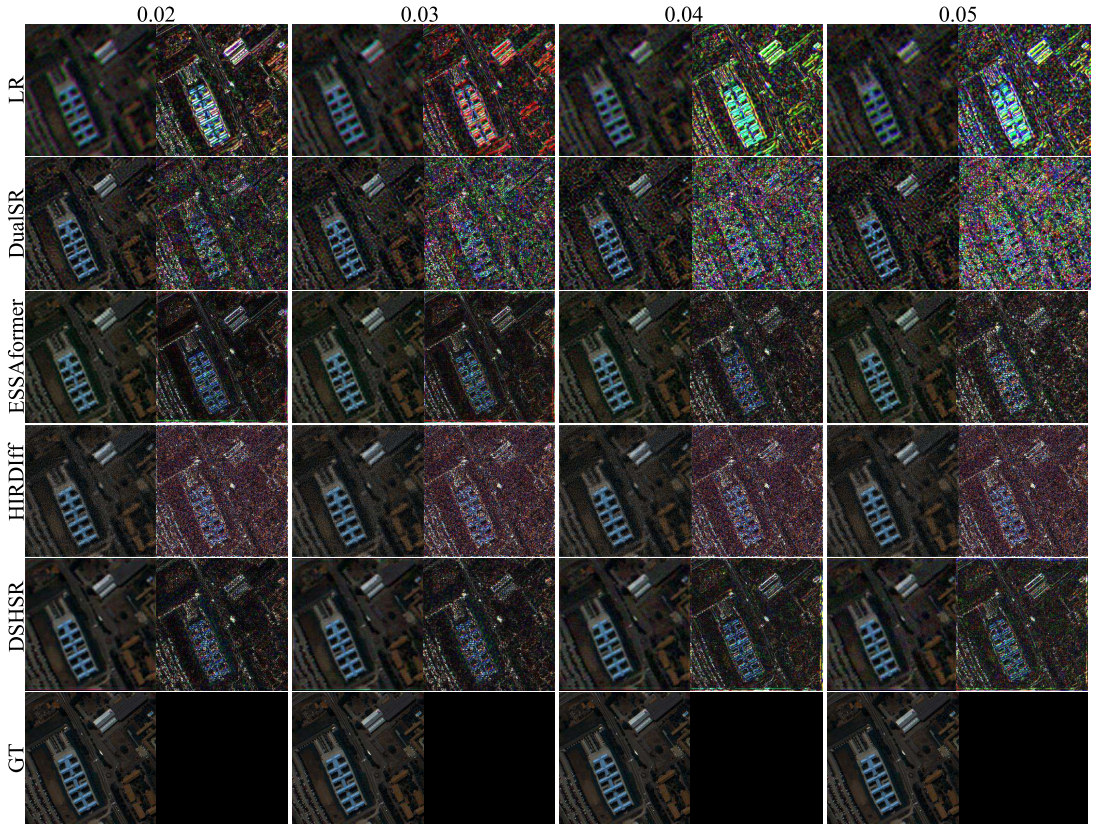
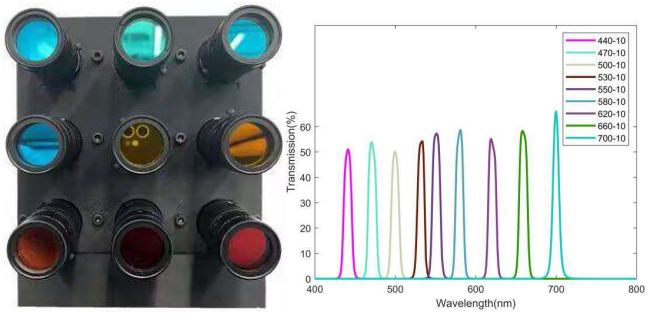


Fig. 13. Test results for data with more severe noise interference in out-of-distribution domain. The Std s of AWGN are set to [0.02, 0.03, 0.04, 0.05].



Lens type	Basler C125-2522-5M-P
Camera type	Hikvision MV-CA004-10UM
Narrowband spectral filter	Thorlabs FB440-10, FB470-10, FB500-10, FB530-10, FB550-10, FB580-10, FB620-10, FB660-10, FB700-10
Image sensor	Sony IMX287

Fig. 14. Transmission rate of all bands and prototype of multispectral array camera. (a) Prototype of a multispectral array camera. (b) Transmission rate curve of all bands. (c) Model specifications of system components.



Fig. 15. Real-world test scene dataset.

images reconstructed by DualSR still exhibit ringing artifacts and jagged edges, as indicated by the white and red arrows. ESSAformer suffers from spectral distortion, which is particularly evident in the synthesized pseudo-color images. It

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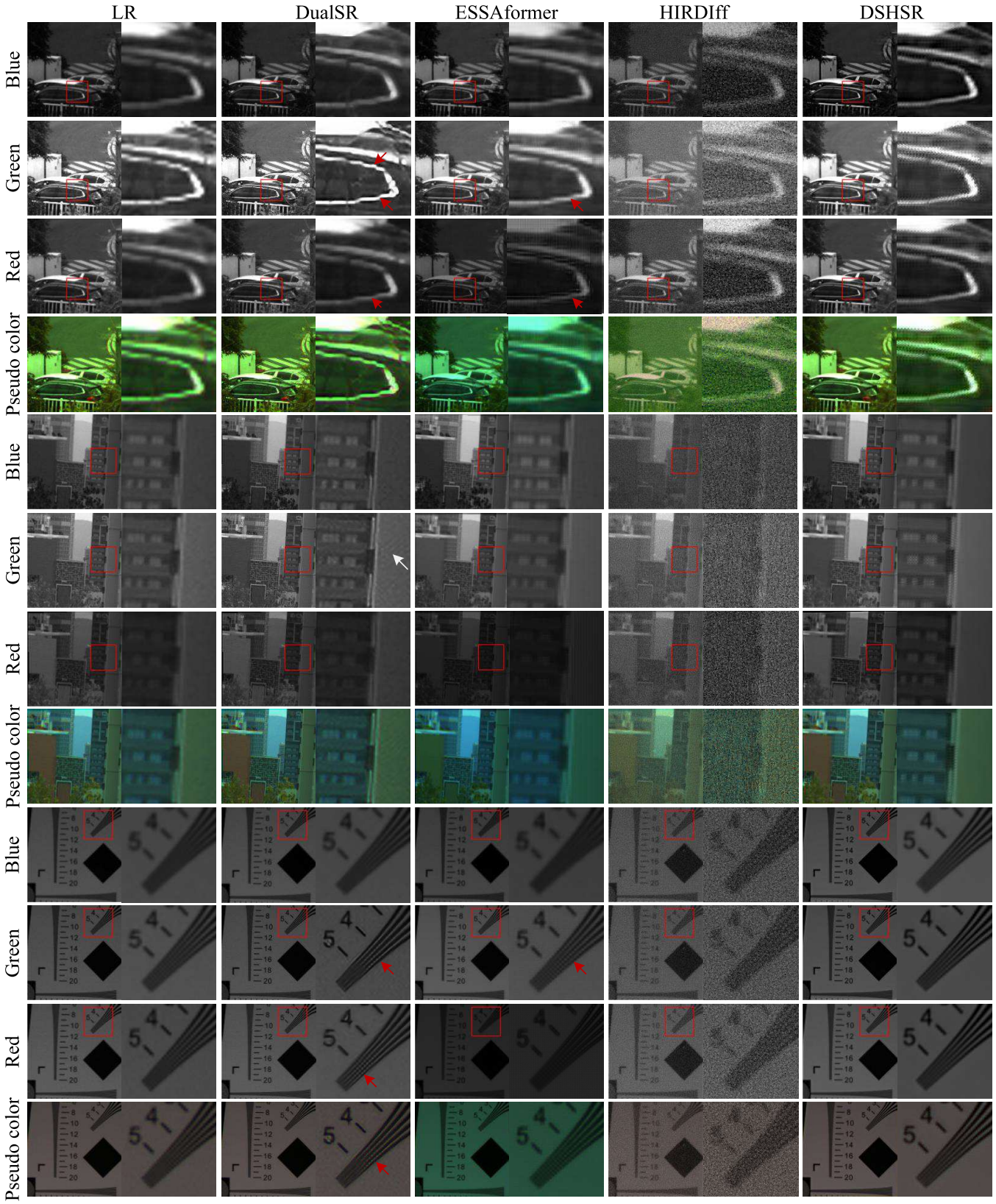


Fig. 16. Reconstruction results comparison on the real multiaperture multispectral datasets.

also displays jagged edge artifacts, as marked by the red arrows. The reconstructed hyperspectral images of HIRDIff are heavily affected by noise, making it difficult to discern target information. In contrast, our proposed method, DSHSR, successfully reconstructs clean images with rich texture details, visually pleasing quality, and high spectral fidelity.

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TABLE III
QUANTITATIVE COMPARISON ON THE REAL DATASETS

	DualSR	ESSAformer	HIRDIff	DSHSR(Ours)
BRISQUE↓	46.15	**44.08**	44.92	**44.72**
NIQE↓	**6.03**	14.14	16.94	**5.46**
SD↓	**0.029**	0.208	0.339	**0.068**

TABLE IV
COMPARISON OF ABLATION EXPERIMENTS ON DIFFERENT LEARNING STRATEGIES AND NETWORK ARCHITECTURES

$\mathcal{L}_1$	$\mathcal{L}_2$	$\mathcal{L}_3$	PSNR↑	SSIM↑
✓			27.86	0.81
✓	✓		27.91	0.82
✓	✓	✓	**28.47**	**0.84**
$\mathcal{S}_1$	$\mathcal{S}_2$		PSNR↑	SSIM↑
✓			27.05	0.79
	✓		27.28	0.80
✓	✓		**28.47**	**0.84**

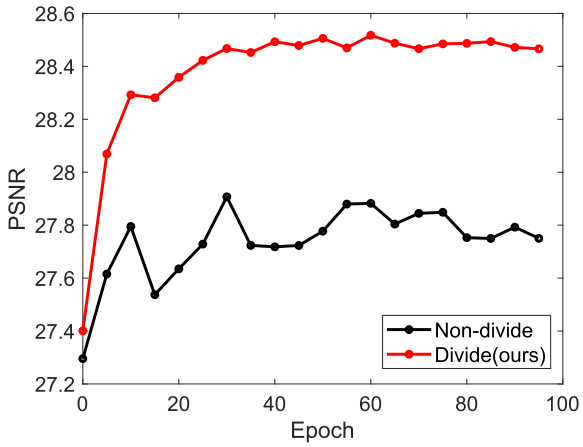


Fig. 17. Convergence analysis of the divide-and-conquer self-supervised learning strategy.

We employ the no-reference image quality assessment metrics BRISQUE and NIQE on both grayscale and color images to evaluate the spatial quality of the reconstructed hyperspectral images. Lower values indicate better spatial quality. As observed in Table III, our method achieves the best NIQE score, while its BRISQUE score is second only to ESSAformer. As shown in the spectral distortion metric of Table III and Fig. 16, ESSAformer exhibits significant spectral distortion in the reconstructed hyperspectral images. The spectral distortion metric from [50] is used to assess the fidelity of the reconstructed spectra. This metric evaluates the similarity between the interband correlation of the LR observed hyperspectral image and that of the reconstructed hyperspectral image. However, it can only measure spectral distortion in low-frequency regions, is highly susceptible to noise interference, and struggles to accurately assess spectral distortion in high-frequency regions.

E. Ablation Study

To validate the effectiveness of the proposed divide-and-conquer self-supervised learning strategy, we trained the DSHSR network using different learning strategies for 30 epochs. In Table IV, $\mathcal{L}_1, \mathcal{L}_2$, and $\mathcal{L}_3$ correspond to (8)–(10) in Section III-D, except that in $\mathcal{L}_1$ and $\mathcal{L}_2$, the training objective is set to the input blurred LR hyperspectral image $\boldsymbol{H}_{\mathrm{BLR}}$. The blur estimation network module is removed, and the SR network is directly constrained by the input blurred LR hyperspectral image. The combination of $\mathcal{L}_1, \mathcal{L}_2$, and $\mathcal{L}_3$ constitutes the proposed divide-and-conquer self-supervised learning strategy. In the ablation experiments, the pretrained FNet from Section IV-B was used, and its parameters were frozen during training. As shown in the quantitative results in Table IV, the proposed learning strategy effectively enhances SR performance.

In Table IV, we included two additional architectural variants to analyze component-level contributions. The first variant, denoted as $S1$, retains only the coarse fusion module. Compared to the original dual-stage SR architecture, this variant removes the fine fusion module and directly applies a 2-D convolution to the output of the Coarse Fusion module to generate the final reconstruction. The second variant, denoted as $S2$, retains only the Fine Fusion module. In this configuration, the coarse fusion module is removed, and the fine fusion module is directly applied to the hyperspectral features extracted by the feature extractor. The proposed attention-driven dual-stage HSR network architecture improves PSNR by 1.42 and 1.19 dB compared to the single-stage architectures that include only the coarse fusion module or only the fine fusion module, respectively. These structural ablation experiments justify the necessity of the proposed dual-stage architecture.

F. Convergence Analysis

We compare two training configurations with and without separating the fusion and blur estimation modules. As shown by the black curve in Fig. 17, when the two modules are nonseparated with $L1$ and $L2$ loss constraining the output of the blur estimation module, the PSNR begins to oscillate around 27.79 dB after 10 epochs, indicating training instability and collapse. In contrast, the red curve in Fig. 17 illustrates the training process under the proposed divide-and-conquer framework, where $L1$ and $L2$ losses supervise the output of the fusion module and $L3$ supervises the output of the blur estimation module. This setting shows a stable convergence trend, with the PSNR steadily improving and reaching 28.52 dB after 60 epochs. These results suggest that separating the fusion and blur estimation modules helps prevent optimization collapse and facilitates more stable training, allowing the fusion module to generate results that are closer to the ground truth.

V. CONCLUSION

In this paper, we propose an aperture array-based self-supervised hyperspectral SR imaging technique, termed DSHSR. We design a lightweight and efficient hyperspectral SR network that simultaneously achieves SR, denoising, and deblurring, along with a noniterative, data-efficient, and ground-truth-free divide-and-conquer self-supervised learning strategy. The proposed DSHSR learning framework

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integrates the forward model with deep neural network priors. The DSHSR network architecture consists of an attention-driven dual-stage SR network and a blur estimation network. To ensure spectral fidelity and effective fusion of high-frequency complementary information across bands, we introduce luminance-texture separation and a band-by-band attention-driven coarse fusion mechanism. This design reduces the parameter burden of 3-D convolutions in the fine fusion stage while maintaining reconstruction performance. The proposed learning strategy effectively leverages both blurred and nonblurred LR hyperspectral images. The blurred LR hyperspectral image serves as input to the network, while the nonblurred LR hyperspectral image acts as a self-supervised signal to constrain the SR network learning. Simultaneously, the blurred LR hyperspectral image serves as a self-supervised signal for the blur estimation network, facilitating accurate blur kernel estimation. As a result, the SR network outputs sharp hyperspectral images. The key advantage of our self-supervised approach is that it guides the network to learn information fusion within bands rather than merely fitting an LR-to-HR mapping, thereby achieving out-of-distribution generalization. Consequently, models trained on simulated data successfully generalize to real-world datasets. Our method reduces runtime by 25× and FLOPs by 9× than DualSR on public remote sensing datasets, while achieving a 1.9 dB PSNR gain. In the optical experiments, DSHSR achieves the best NIQE and the spectral distortion metric that is closely aligned with the spectral response of observed HSI data.

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